Noah Sun

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Education

University of Waterloo – BASc. in Mechatronics Engineering (Candidate) — September 2025 – April 2030

Relevant Coursework: Algorithms and Data Structures, Circuits, Digital Logic — GPA: 3.92 / 4.00

Experience

Full Stack Software Engineer – Thames Valley Financial (Incoming) — September 2026 – December 2026

- Building full stack features for a financial administration platform, including internal tools and product interfaces
- Working AI-forward, steering agentic coding tools toward production-ready features in a small engineering team

Forward Deployed Engineer – Ascendance Foundry — January 2026 – April 2026

- Developed 5 AI workflow automations, introducing input validation and reducing human error by 80%
- Built a GTM automation that uses Python Playwright to find leads and generate outreach emails using Claude API
- Implemented 2 ETL pipelines using Azure Data Factory, delivering per-project statistic dashboards in Power BI

Co-founder and Lead Developer – Elapse | Google Play | App Store | GitHub — June 2024 – August 2025

- Led 4 developers to build an iOS/Android Flutter app serving the 100k+ member VEX Robotics Competition community with real-time schedules, rankings, and match data, receiving 3500+ downloads and a 5.0-star rating
- Designed adaptive match times and scouting forms, increasing team productivity at tournaments by 35%
- Integrated Firebase and Cloud Functions to sync data between team members, reducing scouting times by 30%

Embedded Flight Software Developer – Waterloo Aerial Robotics Group — September 2025 – Current

- Led 4 engineers to develop a 32 MB flash driver using C++ on the STM32 Nucleo to store config details and crash logs
- Built an RTOS-integrated Flash Translation Layer with garbage collection and wear leveling, enhancing flash longevity

V5RC Team Programming Lead – Checkmate Robotics | Demo Video | GitHub — June 2019 – August 2025

- Led team of 6 in robot design and library architecture, winning the Create (innovation) Award at VEX Worlds 2024
- Engineered ArkLib, a modular C++ motion control library focused on scalability and flexibility using OOP, qualifying the team to VEX Worlds in 5 consecutive years and winning the Think (programming) Award at international events
- Designed and tuned PID and control algorithms, achieving 1”/1° precision movements across 300+ matches

Projects

Redact | Live Demo | GitHub

- Built a full-stack face-blurring app (FastAPI, React, Docker) combining RetinaFace detection with OpenCV tracking
- Achieved 97% accuracy by detecting every 5th frame with IoU-tracked fill-in, cutting processing times by 80% to 4 FPS
- Designed dual-signal scene-cut detection and EMA smoothing to keep blur boxes stable across shot changes

Signal - LLM Agent Simulation Platform | GitHub

- Built a multi-layer LLM agent simulation and dashboard in Python and JavaScript with clean separation across a simulation kernel, Claude tool-use harness, and FastAPI/SSE server
- Implemented structured tool-use over Claude API with prompt caching, reducing per-step latency and token cost

CityWatch | GitHub

- Led a team of 4 to build a reporting platform using React Native to analyze 50k+ reports to identify high-risk zones
- Trained a MobileNetV2 road defect classifier with OpenCV preprocessing pipeline, achieving 95%+ accuracy
- Forecasted 30-365 day risk heatmaps from spatial incident patterns and features stored in a PostGIS database

Technical Skills

Languages: C/C++, Python, Java, JavaScript, TypeScript, Dart
Frameworks/AI: ROS2, OpenCV, TensorFlow, Claude API, FastAPI, Express.js, Node.js, React Native, Flutter, Matplotlib
Data/Backend: PostgreSQL, MySQL, Microsoft Azure, Firebase, Pandas
Developer Tools: Git, Docker, Linux, GitHub Actions, Foxglove, REST APIs, SSE, Microsoft Power Apps, MS Office
Hardware: Raspberry Pi, STM32 Nucleo, NVIDIA Jetson Nano, Arduino